CHRIST (DEEMED TO BE UNIVERSITY)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the mini project report entitled **"Mission Simulation Using the Mission Planner Software"** submitted as part of LCA CIA3 for the course **6BTELCS: Drone Technology and its Transformative Applications** being submitted by

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Abstract

This project presents a comprehensive simulation of a drone mission using Mission Planner software integrated with the Software In The Loop (SITL) simulation environment. The simulation demonstrates the complete mission lifecycle of an unmanned aerial vehicle (UAV), encompassing mission parameter configuration, GPS waypoint planning, autonomous flight execution, and controlled landing. By leveraging ArduCopter V4.7.0-dev firmware within a virtual environment, the project validates mission feasibility and system reliability before any real-world deployment. The simulated drone successfully completed a pre-defined triangular flight path, navigating through five sequential mission commands — takeoff, three waypoints, and a landing sequence — with GPS RTK Fixed precision. This exercise underscores the critical role of simulation in modern drone operations, enabling operators to identify and mitigate potential risks, optimise flight paths, and build operational confidence without the cost or danger of live flight testing.

1. Introduction

Unmanned Aerial Vehicles (UAVs), commonly referred to as drones, have rapidly transformed operations across a wide range of industries from precision agriculture and infrastructure inspection to search and rescue operations and aerial surveying. The ability to pre-plan, simulate, and validate drone missions before actual deployment has become an essential practice for safe and efficient drone operations.

Mission Planner is a powerful open-source ground control station (GCS) software developed for ArduPilot-based autopilot systems. It provides drone operators with a comprehensive suite of tools to configure autopilot parameters, design GPS waypoint-based flight missions, perform real-time telemetry monitoring, and simulate complete flight sequences using the Software In The Loop (SITL) environment.

SITL is a simulation framework that allows the ArduPilot firmware to be compiled and run on a standard computer, emulating the behaviour of an actual drone without requiring any physical hardware. When combined with Mission Planner, SITL provides an accurate and safe virtual environment for testing mission logic, evaluating autopilot responses, and training operators.

This project focuses on the end-to-end simulation of a drone mission from initial software setup and parameter configuration through GPS waypoint creation, mission upload, and autonomous flight execution. The practical screenshots captured during simulation provide documented evidence of each stage of the mission, and the results offer meaningful insights into the operational behaviour of the ArduCopter autopilot under autonomous control.

2. Problem Identification

The deployment of UAVs for real-world missions presents several inherent risks and operational challenges. Flying a drone in an unvalidated configuration over unknown terrain without prior testing can result in flyaway incidents, mission failures, property damage, or even injury. Regulatory constraints, limited airspace access, and the high cost of drone hardware further complicate the process of iterative mission testing in live environments.

A key challenge faced by drone operators particularly those new to autonomous flight systems is the difficulty of verifying that a programmed mission will behave as intended once uploaded to the autopilot. Minor errors in waypoint placement, incorrect altitude settings, or misconfigured flight parameters can have serious consequences during real-world execution.

Furthermore, understanding how the ArduCopter firmware interprets and executes mission commands requires hands-on familiarity with the Mission Planner interface and its interaction with the autopilot. Without simulation tools, this learning curve can only be navigated through expensive and risky live flights.

This project addresses these problems by demonstrating how Mission Planner's SITL simulation environment can be used to safely design, upload, and validate a complete drone mission. The simulation allows for iterative refinement of the mission plan in a risk-free virtual setting, building the operator's confidence and technical competence before any real-world deployment is attempted.

3. Objectives

The objectives of this project are as follows:

1. To install and configure Mission Planner software and connect it to the SITL simulation environment for virtual drone testing.
2. To configure mission-critical flight parameters including home location, default altitude, waypoint radius, and loiter radius.
3. To design a multi-waypoint GPS flight path using the Flight Plan (PLAN) tab, defining a structured and safe mission sequence.
4. To upload the configured mission to the simulated ArduCopter autopilot using the Write command and verify correct mission loading via the Read function.
5. To arm the simulated drone, switch to AUTO flight mode, and observe the autonomous execution of the complete mission including takeoff, waypoint navigation, and controlled landing.
6. To document and analyse the drone's behaviour during simulation, including telemetry such as airspeed, ground speed, altitude, battery voltage, GPS status, and flight mode.
7. To evaluate the effectiveness of simulation as a risk-reduction and mission-validation tool in drone operations.

4. Methodologies

The methodology followed in this project consists of six systematic steps, each corresponding to a distinct phase of the mission simulation process.

Step 1: Launch Mission Planner: Mission Planner version 1.3.83 (build 1.3.9384.38258) was launched and connected to the SITL simulation environment using a TCP connection at 115200 baud rate. The connection was established with a TCP5760-1-QUADROTOR virtual vehicle running ArduCopter V4.7.0-dev firmware. The DATA tab confirmed successful connection with real-time telemetry display including heading, altitude, airspeed, battery voltage, and GPS status (RTK Fixed).

Step 2: Configure Mission Parameters: The home location was established at Lat: 12.8631° N, Long: 77.4379° E, ASL: 587.13 m. The default flight altitude was set to 20 metres (relative frame), with a WP Radius of 30 m and Loiter Radius of 30 m. These parameters define the acceptable proximity threshold for waypoint acceptance and circular holding patterns.

Step 3: Create Waypoints: Using the PLAN tab in Mission Planner, GPS waypoints were added interactively on the satellite map to define the drone's triangular flight path. The mission sequence is shown in the table below:

#	Command	Alt (m)	Frame	Latitude	Longitude	Notes
1	TAKEOFF	20	Relative	—	—	Auto ascent
2	WAYPOINT	20	Relative	12.8623447	77.4400212	WP 2
3	WAYPOINT	20	Relative	12.8623565	77.4399340	WP 3
4	WAYPOINT	20	Relative	12.8623185	77.4398670	WP 4
5	LAND	0	Relative	12.8621602	77.4399124	Landing point

Step 4: Upload Mission: The completed mission plan was uploaded to the simulated autopilot by clicking the Write button in the PLAN tab. The mission was verified by reading it back using the Read function to confirm all waypoints were correctly stored in the autopilot's memory.

Step 5: Simulate the Mission: The simulated drone was armed using the Arm/Disarm control in the Actions panel. The flight mode was switched to AUTO, allowing the autopilot to take full autonomous control. The drone began executing the pre-defined mission sequence, progressing through each waypoint in order.

Step 6: Observation and Monitoring: Throughout the simulation, real-time telemetry was monitored via the artificial horizon display, speed indicators, altitude tape, and battery/GPS status bar. The GPS Track and Current Heading indicators were used to verify that the drone was following the correct trajectory. Key telemetry values including airspeed (AS), ground speed (GS), and active flight mode were continuously observed.

5. Proposed Model

The proposed model for this project is a complete software-based drone mission simulation framework built upon the Mission Planner Ground Control Station and the ArduPilot SITL simulation environment. The model integrates several functional components to replicate a real-world autonomous drone mission in a virtual setting.

The core architecture consists of three layers. The first is the Ground Control Station (GCS) layer, represented by Mission Planner, which provides the user interface for mission planning, parameter configuration, telemetry monitoring, and command issuance. The second is the Autopilot Simulation layer, implemented through the SITL framework running ArduCopter firmware, which processes mission commands and simulates vehicle dynamics, sensor inputs, and GPS navigation. The third is the Communication layer, which uses a TCP connection to link Mission Planner with the SITL process, transmitting MAVLink protocol messages bidirectionally.

The mission model follows a sequential state machine approach: the drone begins at a defined home position, receives the arm command and AUTO mode trigger, executes the TAKEOFF command to ascend to the specified altitude, then navigates autonomously through each WAYPOINT in sequence using GPS-guided flight. Upon reaching the final mission item, the LAND command initiates a controlled descent and touchdown at the designated landing coordinates.

This simulation-first model demonstrates that complex autonomous drone behaviours — including GPS waypoint tracking, altitude management, and controlled landing — can be reliably validated using open-source tools. It forms the recommended pre-deployment methodology for any drone mission, regardless of complexity or scale.

6. Results and Discussions

The mission simulation was executed successfully, with the drone completing all planned mission items autonomously. The following results were observed and documented through screenshots captured at each key stage of the simulation.

Screenshot 1: Mission Planner Interface: Active Flight (DATA Tab)

This screenshot shows the Mission Planner DATA tab during active autonomous flight. The HUD confirms the drone is in AUTO mode. Telemetry shows Battery: 12.60V, Current: 28.3A, GPS: RTK Fixed (10 satellites), EKF active, Airspeed and Ground Speed both at 3.5 m/s. The satellite map shows the drone navigating the yellow planned route between waypoints 2 through 5.

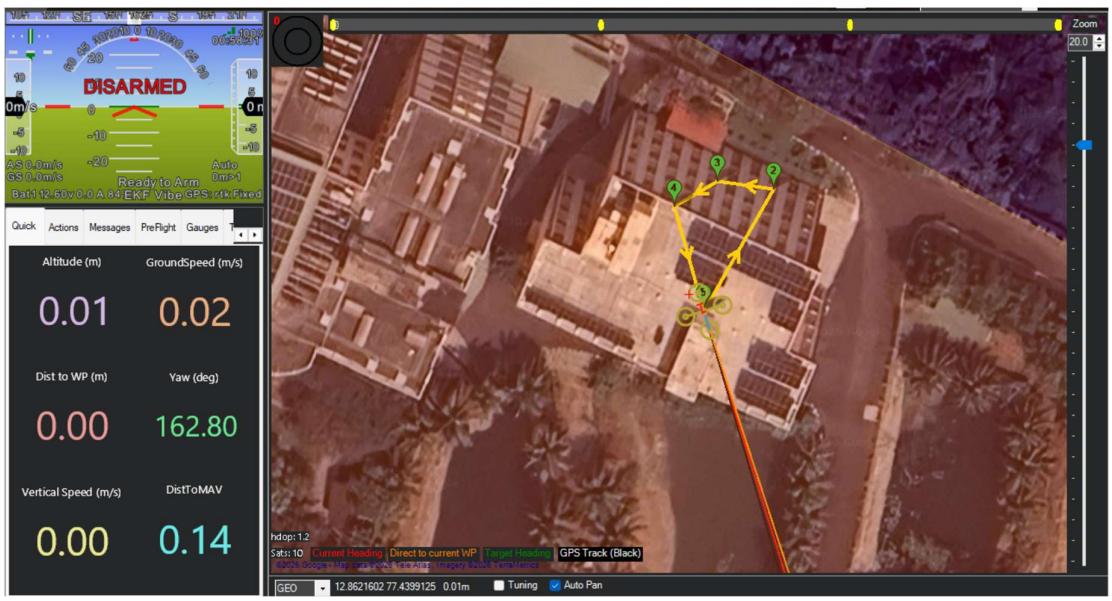

Figure 1: Mission Planner DATA tab:

Screenshot 2: Waypoint Planning: PLAN Tab

The PLAN tab interface was used to design the mission. The satellite map displayed the triangular flight path with yellow connecting lines between waypoints 2 through 5. The waypoint table at the bottom of the screen shows the mission commands including WAYPOINT entries at 25 m and 20 m altitude (Relative frame) and a LAND command at waypoint 5. The Home Location panel confirmed coordinates and ASL altitude of 587.13 m.

Figure 2: Waypoint configuration in Mission Planner

Screenshot 3: Drone Navigating Between Waypoints

This screenshot captures the drone mid-mission in AUTO mode navigating the triangular waypoint path. The purple/magenta GPS track closely follows the intended yellow planned route, confirming accurate GPS-guided tracking. The HUD shows stable horizontal flight at approximately 20–21 metres altitude at 3.0 m/s. GPS remains RTK Fixed with 10 satellites throughout.

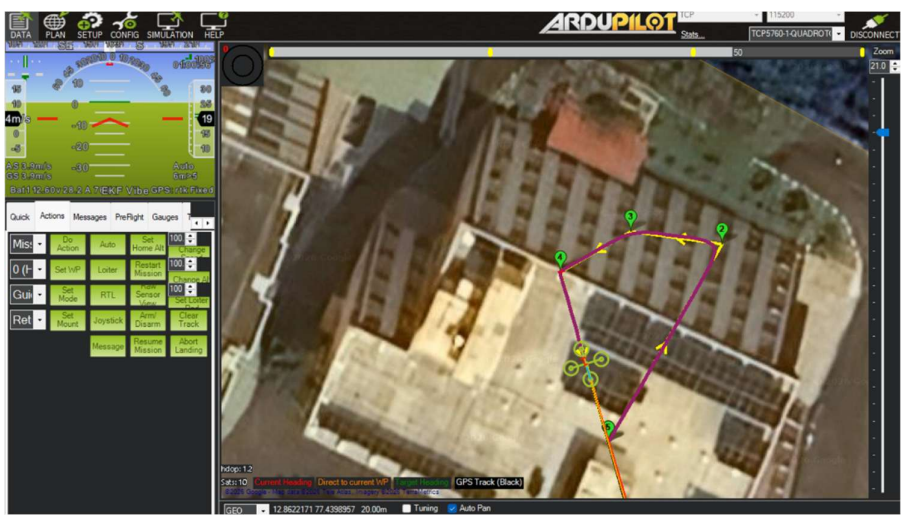

Figure 3: Drone autonomously navigating between waypoints: GPS track (purple) matching planned path (yellow)

Screenshot 4: Mission Complete: Drone Landing at Waypoint 5

The final screenshot shows the drone completing the mission at waypoint 5 (LAND command). The HUD shows airspeed and ground speed at 0.0 m/s confirming touchdown. The mode indicator reads 'Auto 0m>5', confirming the drone has reached waypoint 5. The satellite map shows the purple GPS track completing the full triangular circuit from waypoint 2 through 3, 4, and landing at 5, with battery and GPS status remaining nominal.

Figure 4: Mission successfully completed: drone landed at Waypoint 5, full GPS track visible

Discussion: The simulation results confirm that Mission Planner and SITL together provide a highly effective environment for drone mission validation. The drone followed the pre-programmed GPS waypoint path with high fidelity, as evidenced by the close alignment of the recorded GPS track (purple) with the planned mission path (yellow) on the satellite map. RTK Fixed GPS lock (10 satellites, HDOP 1.2) throughout the mission ensured accurate positioning. Telemetry data remained stable throughout, with no EKF errors, no battery anomalies, and no vibration warnings. The mission was completed fully autonomously from takeoff through landing without any manual intervention, demonstrating the reliability of the ArduCopter autopilot system in a simulated environment. This simulation-first methodology proves to be an indispensable practice for de-risking drone missions before real-world deployment.